

MailSense AI: AI-Powered Sentiment, Risk and Trend Analysis System from Email Correspondence

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Abstract—This study presents the design and implementation of an AI-powered database management system that analyzes sentiment, risk, and trends from email correspondence. The system consists of a normalized Microsoft SQL Server database model, multi-step n8n workflow automations, and an HTML/CSS/JavaScript dashboard interface. The project includes email collection, sentiment analysis, risk detection, category classification, trend calculation, alert notification, weekly reporting, and AI reply suggestion modules. During testing, n8n successfully inserted records into SQL Server tables, stored analysis outputs, generated automatic alerts through trigger logic, and logged workflow executions. This report presents the system architecture, database design, table relationships, workflow processes, test results, and GitHub submission structure in an academic paper format.

Keywords—Database Management Systems, SQL Server, n8n, Artificial Intelligence, Sentiment Analysis, Risk Analysis, Dashboard

I. INTRODUCTION

Corporate email systems contain many critical business signals such as customer requests, complaints, technical problems, pricing questions, and legal risks. Manually reviewing these messages is inefficient and may delay the detection of critical cases. MailSense AI addresses this problem as an integrated system that stores emails in a relational database and transforms them into structured analytical outputs through AI-supported workflows.

The project is developed as a complete DBMS application for the Database Management Systems course. It includes normalized table design, constraints, views, triggers, reporting queries, n8n integration, and a web dashboard interface.

II. PROBLEM DEFINITION

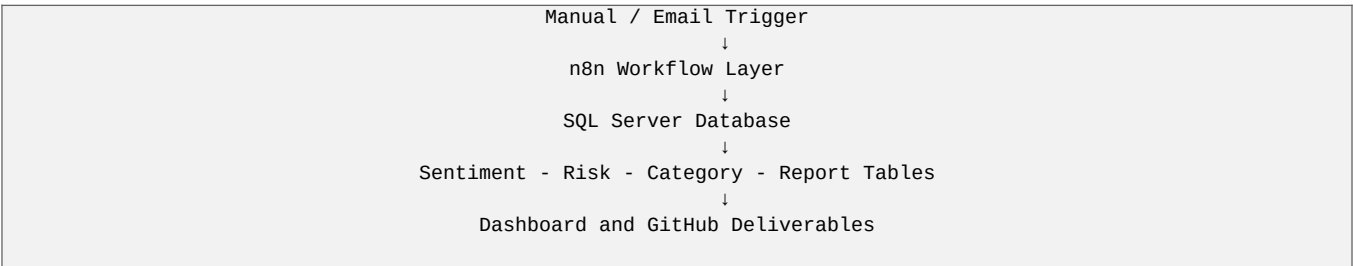
Tracking sentiment trends in email correspondence manually is not sustainable. Customer dissatisfaction, refund requests, cancellation threats, and legal complaints can create financial and reputational risk if they are detected late. Therefore, email content must be stored centrally, analyzed, categorized, and transformed into automatic alerts when necessary.

- Early detection of negative emails.
- Automatic marking of high and critical risk messages.
- Daily and weekly trend reporting.
- Storage of AI reply suggestions as pending drafts.
- Auditable workflow logs for every automation step.

III. SYSTEM ARCHITECTURE

The system consists of four main layers: SQL Server database, n8n automation layer, AI analysis logic, and web dashboard interface. n8n controls the data flow; SQL Server stores persistent data; AI logic generates sentiment and risk scores; and the dashboard presents the results visually.

Fig. 1. MailSense AI system flow.



IV. DATABASE DESIGN

The database design is implemented in Microsoft SQL Server under the name MailSenseAI. The design uses primary keys, foreign keys, unique constraints, check constraints, identity columns, and indexes. These structures maintain data integrity and improve reporting query performance.

TABLE I. Database tables and responsibilities.

Table	Purpose	PK	Relations
departments	Department information	department_id	users
users	System users and roles	user_id	departments, reports
email_accounts	Connected email accounts	account_id	users, emails
email_threads	Email conversation threads	thread_id	emails
emails	Main email table of the system	email_id	sentiment, risk, alerts
sentiment_analysis	AI sentiment analysis results	sentiment_id	emails
risk_analysis	Risk level and recommended action	risk_id	emails, alerts
email_categories	Category dictionary	category_id	email_category_map
email_category_map	Email-category relationship	map_id	emails, categories
keywords / email_keyword_map	Keyword relationships	keyword_id	emails
alerts	Critical case alerts	alert_id	emails, risk_analysis
reports	Daily/weekly reports	report_id	users
ai_response_suggestions	AI reply suggestions	suggestion_id	emails
workflow_logs	n8n execution logs	log_id	emails

The email_category_map table was added to store the relationship between emails and the category dictionary. This update was also included in schema.sql so that the category workflow works correctly in a clean installation.

V. SQL FILE STRUCTURE AND DATA INTEGRITY

Project SQL files are separated under the database folder. schema.sql creates tables and constraints; seed.sql inserts sample data; queries.sql contains reporting queries; views.sql creates summary views; and triggers.sql defines automatic business rules.

TABLE II. SQL and project file purposes.

File	Description
database/schema.sql	Creates database schema, tables, constraints, and indexes.
database/seed.sql	Inserts demo users, emails, analysis outputs, alerts, and logs.
database/queries.sql	Contains reporting and validation queries.
database/views.sql	Creates analytical views for summaries and dashboards.
database/triggers.sql	Creates automatic alert and workflow log triggers.
workflows/*.json	Exported n8n workflow definitions.
frontend/index.html, style.css, app.js	Static dashboard interface for visual reporting.

The trg_create_alert_on_high_risk trigger automatically inserts alert records when a high or critical risk record is inserted into risk_analysis. This ensures that the business rule is protected at the database level, not only at the n8n level.

VI. N8N WORKFLOW DESIGN

Eight separate workflows were prepared in n8n and exported as JSON files into the workflows folder. Each workflow represents a specific business process. This modular structure makes testing and debugging easier.

TABLE III. n8n workflow modules and test outputs.

No	Workflow	Function	Tables	Output
01	Email Collector	Collects a test email and stores it in the emails table.	emails, workflow_logs	Successful insert and log
02	AI Sentiment Analysis	Generates sentiment score and label from email text.	sentiment_analysis, emails	Negative/positive/neutral result
03	Risk Detection	Produces risk level, risk score, and recommended action.	risk_analysis, alerts	High/critical risk detection
04	Category Classification	Maps the email to a business category.	email_category_map	Complaint category
05	Trend Calculation	Creates an overall sentiment and risk trend report.	reports	Daily report record
06	Alert Notification	Marks pending alerts as sent.	alerts	Alert marked as sent
07	Weekly Report Generator	Generates a weekly summary report.	reports	Weekly report record
08	AI Reply Suggestion	Generates a reply suggestion for risky emails.	ai_response_suggestions	Pending reply suggestion

The first workflow inserts a test email into the emails table. The second workflow stores sentiment results. The third workflow calculates risk and triggers alerts. The fourth workflow maps categories through email_category_map. The fifth and seventh workflows generate reports. The sixth workflow marks pending alerts as sent. The eighth workflow stores AI reply suggestions.

VII. AI ANALYSIS AND BUSINESS RULES

During the demo stage, rule-based AI logic was used. When expressions such as refund, delayed, unhappy, angry, cancel, or legal complaint are detected, sentiment and risk scores are increased. This approach validates the database and workflow flow before integrating a real AI API.

TABLE IV. Analysis rules and sample outputs.

Rule	Result	Sample Value
refund, delayed, unhappy	negative sentiment	sentiment_score = -0.82
negative or refund	high risk	risk_score = 0.82
legal complaint, cancel, angry	critical risk	risk_score = 0.96
risk high/critical	alert generation	alert_status = pending/sent
risky email	AI reply suggestion	approval_status = pending

VIII. TEST RESULTS

During testing, the n8n to SQL Server connection was configured using the n8n_user SQL login, TCP/IP, and port 1433. Each workflow was executed with Execute Step and the outputs were verified in SQL Server Management Studio.

TABLE V. Application test scenarios.

Test	Purpose	Target	Observed Result	Status
T1	n8n email insert	emails table	email_id=6 created	Success
T2	Sentiment analysis	sentiment_analysis	negative, -0.82	Success
T3	Risk detection	risk_analysis + alerts	high risk, 0.82 and alert	Success
T4	Category classification	email_category_map	Complaint category	Success
T5	Trend report	reports	daily report record	Success
T6	Alert notification	alerts	pending -> sent	Success
T7	Weekly report	reports	weekly report record	Success
T8	AI reply suggestion	ai_response_suggestions	pending suggestion	Success

IX. WEB DASHBOARD AND GITHUB SUBMISSION

The dashboard in the frontend folder displays the total number of emails, negative email count, risky emails, sentiment distribution, keywords, system alerts, and workflow logs. It was developed using HTML, CSS, and JavaScript and can be opened directly through index.html without additional installation.

The GitHub submission structure includes README.md, README_TR.md, database, workflows, docs, reports, frontend, and assets folders. All eight JSON files in the workflows folder are real exported n8n workflow files and are not empty.

TABLE VI. Submission folders and contents.

Folder	Content	Submission Role
database	schema, seed, queries, views, triggers	Recreates and tests the database
workflows	8 exported n8n JSON files	Allows automation import
docs	Turkish/English academic reports	Academic submission documents
reports	Sample weekly reports	Shows system output
frontend	index.html, style.css, app.js	Runs the dashboard interface

X. SECURITY, LIMITATIONS AND FUTURE WORK

Real email passwords, API keys, and connection credentials must not be uploaded to GitHub. Therefore, `.env.example` is provided only as a template. In a real deployment, secrets should be stored in a local `.env` file and protected through `.gitignore`.

Future improvements may include real Gmail/Outlook integration, OpenAI API integration, user authentication, real-time dashboard data, and advanced charts. The workflows can also be scheduled to run automatically using schedule triggers.

XI. CONCLUSION

MailSense AI was completed as a working database management system that performs sentiment, risk, and trend analysis from email correspondence. SQL Server database tables, n8n workflow automations, triggers, reporting queries, and the web dashboard work together. The project demonstrates data modeling, relational integrity, automation, analysis, and reporting concepts in an integrated DBMS scenario.

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